# A Layered Risk-Signal Architecture for Fraud-Aware Job Search: From Job Post and Employer to Conversation and Ranking

***Abstract*—Recruitment fraud reaches a job seeker in three places: the advertisement, the employer behind it and the private conversation that follows. Existing tools cover one place each, and none of them changes which jobs the seeker is shown. This paper presents a four-component architecture in which three detectors each return a risk score, and a fourth component turns these scores into a ranked job list. The detectors are a multilingual (English and Sinhala) job post classifier, an explainable employer legitimacy verifier that fuses registry, reputation and website evidence with a Random Forest score, and a conversation detector that identifies four manipulation tactics. The ranking component aggregates the risks with the Entropy Weight Method, and combines safety with TF-IDF skill relevance using TOPSIS. The post detector reaches 88.5% accuracy and a ROC-AUC of 0.924, the conversation detector reaches 85.3% binary accuracy and a macro F1 of 0.83 over four tactics, and the employer verifier is evaluated on a held-out test split with a single evaluation: its explainable weights catch 78.1% of fraudulent employers and a data-driven weighting 93.3%, which measures the price of explainability. The ranking component keeps relevance unchanged at Precision@10 of 0.616 and lowers the average risk of the Top-10 list from 0.101 to 0.090, and the unsafe rate from 2.80% to 2.40%.**

***Keywords*—online recruitment fraud, job recommendation, multilingual classification, employer verification, explainable AI, entropy weight method, TOPSIS**

## I. Introduction

Online recruitment fraud causes financial loss, identity theft and distress for job seekers [1]. A typical scam has two phases: a deceptive advertisement attracts the victim, and the exchange then moves to private messages where pressure is used to obtain money or personal data [2]. Automated work has mostly addressed the first phase as text classification [1], [3]. Other work shows that employer information [4], [5] and message content [6] are useful signals, but each study treats its signal on its own. Recommenders add a further gap: they rank jobs by skill match [7] and never use fraud evidence, so a fraudulent vacancy can still appear first.

The problem is sharp in Sri Lanka. Job seekers meet offers in English and Sinhala on classified sites, poster images and social media, pyramid-style schemes are reported in the news, and official registers exist but are scattered over several portals. A tool for this market must read both languages, use local sources and explain its verdict to a non-expert. We therefore ask whether post, employer and conversation signals can be produced by independent detectors that share one simple output contract, how the three risks should be combined with each other and with skill match without hand-set weights, and what it costs in detection quality to choose employer fusion weights that a user can understand.

This paper treats these as one problem and makes three contributions. The first is a risk-signal architecture in which three independent detectors (post, employer, conversation) share one output contract, a risk in $[0, 1]$ plus a human-readable explanation, so that each can be built, evaluated and replaced separately. The second is a ranking method that aggregates the three risks with the Entropy Weight Method (EWM) into a safety score and ranks jobs with TOPSIS on safety and skill relevance. The third is a held-out study of the employer verifier, with a single test evaluation, intervals and a measured cost of choosing explainable fusion weights, together with component-level results for the post and conversation detectors. Section II reviews related work, Section III gives the architecture, Section IV the components, Section V the evaluation, and Section VI the limitations and conclusion.

## II. Related Work

The EMSCAD corpus of Vidros et al. [1] is the usual benchmark, and classical classifiers on it, such as Random Forest, reach high accuracy [3]. Metadata of the advertisement adds further signal [5]. These studies use static English text, and Sinhala postings, common in Sri Lanka, are not covered.

Employer-level evidence has also been studied. Mahbub et al. added registry and website features to a Random Forest and raised accuracy from 89.22% to 91.86% [4]. Pangare et al. combined identity checks and review data with text classification [8]. These systems use the registries of other countries, give no account of each source's contribution, and do not report how combination weights were chosen apart from the test data.

For conversations, BERT-based models detect scam messages better than keyword filters [6], and OCR can recover text from screenshots [9]. Anagha et al. described four manipulation tactics in job scams, urgency, fear of missing out,

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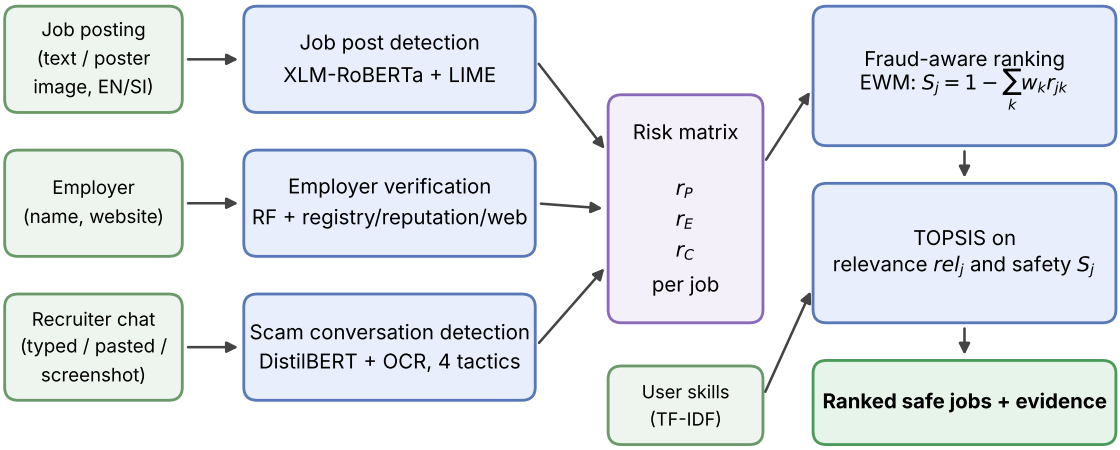



Fig. 1. Architecture: three detectors produce risk scores that the ranking component aggregates and combines with skill relevance to rank jobs.

sunk cost and social proof, but not as an automated detector [2].

For recommendation and ranking, TF-IDF with cosine similarity gives content-based job recommendation [7]. Risk has been added as layoff risk [10], not fraud, and entropy weighting with TOPSIS has been used for transparent ranking with occupational criteria [11]. Fraud-oriented features have been used for classification only [12].

No reviewed work connects post, employer and conversation signals to the ranking a job seeker sees.

## III. System Architecture

Fig. 1 shows the architecture. A job posting goes to the post detector. The employer named in it goes to the employer verifier. If the seeker talks to the recruiter, the message, typed, pasted or taken from a screenshot, goes to the conversation detector. Each component returns a risk $r \in [0, 1]$ (higher is riskier) and an explanation: token-level attributions for the post and conversation detectors, and an evidence report for the employer verifier. The employer risk is $r_E = 1 - S_B/100$, where $S_B$ is the legitimacy score of the verifier. The ranking component reads the risks stored for each job and the seeker's skills and returns the ranked list. The only coupling between components is this contract, so a component can be retrained or replaced without changing the others. A missing signal, for example a job with no conversation yet, is left out of the risk matrix and its indicator is not used for that job.

Four choices shape the architecture. First, each detector has its own data, model and service, so a failure or retraining in one does not affect the others. Second, a risk number alone does not tell a job seeker what to do, so each detector also returns the evidence behind it. Third, when evidence is missing, the output is marked as lower confidence and is not silently treated as safe. Fourth, the risks are merged by a data-driven method and not by weights fixed by hand.

TABLE I. Data used by the trained components

| Comp. | Data | Split |
|---|---|---|
| Job posts | 4,590 posts (3,698 legit., 892 fraud), EN and SI | Stratified 80/20 |
| Employers | 1,514 employers (978 legit., 536 fraud) | Domain-grouped 60/20/20 (901/313/300) |
| Messages | 7,433 messages (binary); 3,840 messages (tactics) | Group-aware split; test 744 and 758 |

## IV. Components

This section describes the four components in the order of the pipeline. For each one it gives the data, the model and the output that is passed on as a risk. Table I summarises the data of the three trained components. The fourth component, the ranking component, needs no training data, because it aggregates the outputs of the other three with unsupervised methods.

The three splits differ because the data differ: the job posts are independent and can be split at random within each class, while the employer and message data contain records that share a website or a source conversation, and a random split would leak information between training and test. This is why those two use grouped splits. The post split is stratified and may therefore be slightly optimistic if near-duplicate posts exist, which is a further reason to treat its result with care.

### A. Multilingual job post detection

The corpus has 4,590 posts (3,698 legitimate, 892 fraudulent) in English and Sinhala. Local posts were collected from ikman.lk, poster-style images from topjobs.lk were converted to text with a vision model, and 866 of the fraudulent posts come from EMSCAD [1]. Language is identified from the share of Sinhala Unicode characters (U+0D80–U+0DFF, above 30%). Labels were assigned by a rule-guided automated labeller using fraud indicators (upfront fees, unrealistic

pay for unskilled roles, missing company name, personal email domains), not by independent human annotators. XLM-RoBERTa [13] was fine-tuned on a stratified 80/20 split with inverse-frequency class weights. LIME [15] is the primary explanation method and gives the most fraud-like and legitimate-like tokens for each post, and a leave-one-out token attribution, which removes one token at a time and measures the change in fraud probability, is used as a check. The model is served through a FastAPI back end that returns the predicted class, its confidence and the class probabilities; for poster images the vision model first extracts the text, which then follows the same path. The fraud probability is used as the post risk $r_P$.

### B. Employer legitimacy verification

For the employer in a post, three layers gather evidence in parallel. The *registry layer* searches Sri Lankan sources (Colombo Stock Exchange, eROC/DRC, Central Bank lists, Board of Investment, insurance regulator, Specified Business Enterprises list), expands brand names to legal names and records a trace. The *reputation layer* searches review and social platforms and detects scam reports. The *website layer* checks HTTPS, About, Contact, privacy and terms pages, and payment and urgency wording. A Random Forest [17] on 14 structured features (200 trees, depth 8, balanced classes; scikit-learn [18]) gives the model score $P_{ML}$. The legitimacy score is the weighted combination of classifier outputs [19]

$$S_B = 100\,(w_1 P_{ML} + w_2 R_{reg} + w_3 R_{rep} + w_4 R_{web}), \quad (1)$$

with deployed weights $(0.40, 0.30, 0.20, 0.10)$, where each $R$ is a layer score in $[0, 1]$. Scores of 70 or more are Low risk, 45–69 Medium, below 45 High. A missing registry match earns no points but is not treated as fraud, because many genuine small enterprises are unlisted. The three layers run concurrently and their results are cached, and a failed layer lowers the confidence of the answer without stopping it; a low-confidence flag is also returned when little information is supplied or a website cannot be scanned. The explanation is an evidence report with the registry matches, platforms found, website findings, per-layer points and plain-language reasons, not a post-hoc attribution.

### C. Scam conversation detection

Phase 1 is binary (scam or legitimate) on 7,433 messages that combine EMSCAD fraud postings, 322 messages collected from Reddit, LinkedIn and Facebook, and team-authored, reviewed conversation samples. Phase 2 assigns the four tactics of [2] as a multi-label task on 3,840 group-deduplicated messages, with a group-aware split to keep variants of one conversation together. DistilBERT [16] was fine-tuned for Phase 1, and its encoder was then reused for Phase 2 with a multi-label head and per-tactic thresholds tuned on validation data. Screenshots are read with EasyOCR, with Tesseract as fallback, and the text passes through the same classifiers. Word-level SHAP [14] and LIME attributions are mapped to plain-language warnings, so the user sees which phrases and which tactic triggered the alert. The service runs behind FastAPI with Socket.IO for live chat, and accepts typed, pasted and screenshot input. The scam probability is used as the conversation risk $r_C$.

TABLE II. ILLUSTRATIVE RANKING EXAMPLE (HYPOTHETICAL INPUTS)

| Job | $rel$ | $r_P$ | $r_E$ | $r_C$ | $S_j$ | $C_j$ | Rank |
|---|---|---|---|---|---|---|---|
| 1 | 0.82 | 0.10 | 0.08 | 0.05 | 0.92 | 0.998 | 1 |
| 2 | 0.78 | 0.85 | 0.70 | 0.40 | 0.35 | 0.248 | 5 |
| 3 | 0.74 | 0.20 | 0.35 | 0.10 | 0.80 | 0.755 | 3 |
| 4 | 0.66 | 0.05 | 0.15 | 0.05 | 0.92 | 0.774 | 2 |
| 5 | 0.60 | 0.15 | 0.10 | 0.60 | 0.71 | 0.531 | 4 |

### D. Fraud-aware ranking

User skills and job descriptions are TF-IDF vectors [20], and the cosine similarity gives $rel_j \in [0, 1]$.

For $n$ candidate jobs the risk matrix is $X = [r_{jk}]$, $k \in \{P, E, C\}$ for the post, employer and conversation risks. With $p_{jk} = r_{jk} / \sum_j r_{jk}$, the entropy of indicator $k$ is $e_k = -\frac{1}{\ln n} \sum_j p_{jk} \ln p_{jk}$, its weight $w_k = (1 - e_k) / \sum_m (1 - e_m)$ [21], and

$$R_j = \sum_k w_k\, r_{jk}, \qquad S_j = 1 - R_j. \quad (2)$$

An indicator that varies more across jobs carries more information and gets more weight, so the weights are objective but depend on the candidate set.

Jobs are then ranked with TOPSIS [22], which takes $rel_j$ and $S_j$ as benefit criteria, normalises and weights the columns, finds the ideal and anti-ideal solutions, and ranks jobs by the closeness $C_j = d_j^- / (d_j^+ + d_j^-)$, so a job rises only if it is both relevant and safe. Each result shows relevance, safety, closeness, matched skills and suggested skills, with the evidence reports of the detectors.

### E. Worked example of the ranking

To make the method concrete, Table II applies it to five hypothetical jobs. The risks and relevance values are chosen for illustration and are not experimental data. Job 2 has high relevance (0.78) but a high employer risk (0.70), a high post risk (0.85) and a mid conversation risk. The entropy weights are $w = (0.39, 0.26, 0.35)$ for $(P, E, C)$, and equal TOPSIS weights of 0.5 are used for relevance and safety. By relevance alone the order is $1, 2, 3, 4, 5$. After fraud-aware ranking it is $1, 4, 3, 5, 2$: job 2 falls from second place to last, while job 4, which is less relevant but very safe, rises from fourth to second. Job 5 has the lowest relevance and a high conversation risk, so it stays near the bottom but still ranks above job 2. This shows the intended behaviour: a relevant but unsafe job is demoted and a safe job is promoted.

## V. IMPLEMENTATION

All four components are implemented in Python. The three detectors run as separate FastAPI services, so each can be deployed and scaled alone, and a React Native mobile application calls them through a common interface. The post detector returns a class, confidence and token attributions, the

employer verifier returns a score, band and evidence report, and the conversation detector returns a binary decision, tactic labels and warnings. The ranking component reads the stored risks and the user's skills, and returns the ranked list. The employer verifier queries public registries and review pages over the network, so it runs its layers in parallel, caches repeated lookups and times out slow sources; a slow or blocked source lowers the confidence flag and does not block the answer. The settings that decide the reported results are fixed so that the studies can be repeated. For the employer verifier, the domain-grouped split uses seed 42, the Random Forest settings are the ones given in Section IV-B, and the five weight profiles, the grid search and the threshold sweep are run from the same code on the validation split before the single test evaluation. For the conversation detector, the checkpoint with the best validation F1 is kept, and the per-tactic thresholds of the conversation detector are tuned on validation data only. The test sets of all three components are not used for any choice of model, weight or threshold.

A typical use shows how the parts work together. A job seeker opens a poster or posting, and the post detector returns a fraud probability with the words that drove it. The employer named in the post is sent to the employer verifier, which returns a band and an evidence report that the seeker can check on the official registers. If the recruiter then writes to the seeker, the message or a screenshot of it goes to the conversation detector, which returns a warning and the manipulation tactic. These three risks are stored with the job, and the ranking component uses them the next time the seeker's job list is ranked, so a job with a suspicious employer or a pressuring recruiter moves down the list without the seeker having to judge each signal alone.

## VI. Evaluation

The four components have been integrated and run together in the mobile application, so a job passes through detection, verification and ranking in one flow. The quantitative evaluation below is reported for each component on its own data, because the components were trained on different datasets and no single labelled set covers all of them.

### A. Job post and conversation detectors

Table IV summarises the two classifiers. On the 635-post test set (479 legitimate, 156 fraudulent), the post detector reaches 88.5% accuracy and a ROC-AUC of 0.924, with a fraud-class F1 of 0.77. Training accuracy is 96.5%, a gap of 8 points that is consistent with the shift from Western English training posts to Sri Lankan test posts. The conversation detector reaches 85.3% binary accuracy on 744 held-out messages (491 legitimate, 253 scam). For the four tactics on 758 held-out messages it reaches a macro F1 of 0.83 and a micro F1 of 0.87, with F1 of 0.93 (urgency), 0.91 (social proof), 0.86 (sunk cost) and 0.64 (fear of missing out). The last is weakest because of low recall (0.54) and varied wording.

TABLE III. Tactic classification (Phase 2) on 758 held-out messages

| Tactic | Prec. | Rec. | F1 | Support |
|---|---|---|---|---|
| Urgency pressure | 0.906 | 0.954 | 0.929 | 241 |
| Social proof | 0.915 | 0.905 | 0.910 | 179 |
| Sunk-cost influence | 0.866 | 0.853 | 0.859 | 68 |
| Fear of missing out | 0.775 | 0.545 | 0.640 | 101 |

TABLE IV. Component results on held-out test data

| Detector | Task (model) | $n$ | Key result |
|---|---|---|---|
| Post | Post (XLM-R) | 635 | Acc. 0.885, AUC 0.924 |
| Conversation | Message (DistilBERT) | 744 | Acc. 0.853 |
| Conversation | 4 tactics (DistilBERT) | 758 | Macro F1 0.835, micro 0.872 |

The confusion matrix of the post detector on the test set has 439 true legitimate, 123 true fraudulent, 40 false-positive and 33 false-negative posts. The fraud class is the harder one (precision 0.75, recall 0.79) because it is the minority class and its Sri Lankan examples are few. On unseen Sri Lankan poster images, the model classified a legitimate corporate post as legitimate with 98% confidence and a Sinhala-language multi-level-marketing recruitment post as fraudulent with 99% confidence, which supports the cross-lingual transfer argued for in Section IV-A. Classical models on EMSCAD give F1 of 0.79–0.82 [1], but these use different data, so the comparison is indicative only. For the conversation detector, no earlier system detects tactics in recruitment conversations, so no direct baseline exists, and the binary result is of the same order as BERT-based SMS and phishing detectors [6].

Exact-match accuracy of the tactic labels is 85.4% and the Hamming loss is 0.049.

### B. Employer verification protocol

The labelled set has 1,514 employers (978 legitimate, 536 fraudulent). Fraud labels come from Sri Lankan news reports of banned schemes and from public scam reports on social media, and website evidence was compiled manually from public sites. The split is by website domain: training 901, validation 313 and test 300 (195 legitimate, 105 fraudulent), seed 42. The model is trained on the training split only, weights and thresholds are chosen on the validation split, and the frozen design is evaluated once on the test split. Five weight profiles and a grid search (step 0.05) were compared, thresholds from 20 to 80 were swept, and cost ratios of 3:1, 5:1 and 10:1 for missed fraud against false alarms were used [23]. Intervals are Wilson 95% intervals [24]. "Fraud recall" is the share of fraudulent employers scored below the cut-off, and "FPR" the share of genuine employers wrongly flagged.

### C. Employer verification results

Table V gives the profiles on the test split at cut-off 45. The explainable weights reach 0.870 accuracy and 0.781 fraud recall. The ML-dominant weights and the grid-search weights (0.50/0.10/0.25/0.15, chosen on validation) reach recall of 0.876 and 0.933. Giving the registry most of the weight, or dropping reputation, lets more than half of the fraud pass

TABLE V. EMPLOYER WEIGHT PROFILES ON THE TEST SPLIT (CUT-OFF 45, $n = 300$)

| Profile (weights) | Acc. | F1 | Fraud rec. | FPR |
|---|---|---|---|---|
| Explainable (.40/.30/.20/.10) | 0.870 | 0.902 | 0.781 | 0.082 |
| ML-dominant (.70/.10/.10/.10) | 0.920 | 0.939 | 0.876 | 0.056 |
| Registry-heavy (.20/.60/.10/.10) | 0.733 | 0.814 | 0.429 | 0.103 |
| Equal (.25 each) | 0.850 | 0.887 | 0.743 | 0.092 |
| No reputation (.50/.40/0/.10) | 0.773 | 0.846 | 0.429 | 0.041 |
| Grid best (.50/.10/.25/.15) | 0.930 | 0.945 | 0.933 | 0.072 |

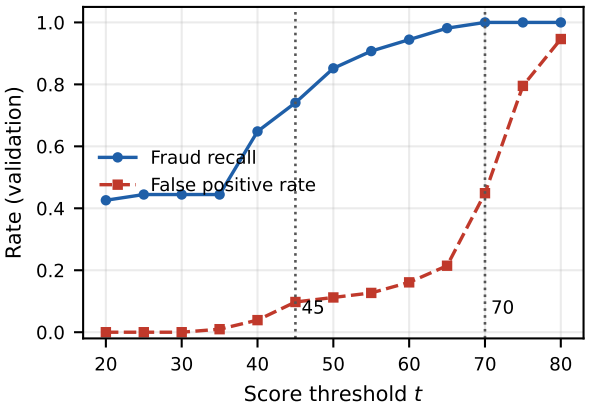



Fig. 2. Fraud recall and false positive rate against the score threshold (validation split), with the deployed bands at 45 and 70.

(recall 0.429 in both), so registry evidence alone is not enough. The price of the explainable weights is about 15 points of fraud recall at a similar false positive rate (0.082 against 0.072).

Fig. 2 shows the validation sweep, and the cost-optimal threshold is 65 for all three cost ratios. Table VI gives the risk bands on the test split. With the deployed bands (45 and 70), none of the 107 employers in the Low band is fraudulent, and the High band holds 82 of 105 fraudulent employers (78.1%, Wilson interval 69.3–84.9%), with 16 genuine employers in High and 72 in Medium. The bands chosen by the validation rules (55 and 60) put more fraud in High (96 of 105, 91.4%, interval 84.5–95.4%) but let 5 fraudulent employers into Low. The threshold position is therefore a choice about which error is acceptable, and the study makes it visible.

### *D. Ranking protocol and results*

The ranking component is evaluated against a TF-IDF and cosine-similarity baseline that ranks by relevance only, using a ranked list of ten jobs per job-seeker profile, with risks supplied by the three detectors. Relevance is measured by Precision@10, AP@10, MRR@10 and NDCG@10, and safety by the average risk, the average safety and the unsafe rate in the Top-10 list, where the unsafe rate is the share of listed jobs that are fraudulent or high-risk. Precision@10 is the share of relevant jobs in the list, AP@10 averages precision at each relevant position, MRR@10 is the inverse rank of the first relevant job, and NDCG@10 rewards relevant jobs placed higher [25]. Both methods rank the same candidate set, so any difference comes from the risk terms alone. Table VII gives the results.

TABLE VI. EMPLOYER RISK BANDS ON THE TEST SPLIT (LEGITIMATE / FRAUDULENT)

| Bands | High | Medium | Low |
|---|---|---|---|
| Deployed 45/70 | 16 / 82 | 72 / 23 | 107 / 0 |
| Validation-selected 55/60 | 29 / 96 | 13 / 4 | 153 / 5 |

TABLE VII. RANKING RESULTS AT $K = 10$

| Metric | Baseline | Proposed | Change | Better |
|---|---|---|---|---|
| Precision@10 | 0.6160 | 0.6160 | 0 | Same |
| AP@10 | 0.8612 | 0.8627 | +0.0015 | Proposed |
| MRR@10 | 1.0000 | 1.0000 | 0 | Same |
| NDCG@10 | 0.9648 | 0.9655 | +0.0007 | Proposed |
| Average Risk@10 | 0.1006 | 0.0900 | −0.0106 | Proposed |
| Average Safety@10 | 0.8994 | 0.9100 | +0.0106 | Proposed |
| Unsafe Rate@10 | 2.80% | 2.40% | −0.40 pp | Proposed |

The relevance metrics are unchanged or slightly higher: Precision@10 and MRR@10 are identical, and AP@10 and NDCG@10 rise by 0.0015 and 0.0007. The safety metrics improve. The average risk of the Top-10 list falls from 0.1006 to 0.0900 (about 11% lower), and the unsafe rate falls from 2.80% to 2.40%. Fraud-aware ranking therefore lowers the risk shown to the seeker without losing relevance.

## VII. DISCUSSION

The ranking results show how the three risk signals change what a seeker sees. Relevance is kept: Precision@10 and MRR@10 are identical to the baseline, so the best match is still listed first. At the same time the average risk of the list falls by 0.0106, an 11% relative drop, and the unsafe rate falls from 2.80% to 2.40%, a 14% relative drop.

The three detectors were built, trained and evaluated separately and meet only at the risk-and-explanation contract, which suggests that the contract is enough for integration. This has been shown at the design level and by the worked example, but not yet on joint data. EWM removes hand-set risk weights and TOPSIS balances risk and relevance, but the weights follow the spread of the data, so an indicator that is nearly constant in a candidate set gets almost no weight even when it matters. For the employer verifier, understandable weights cost about 15 points of fraud recall at the deployed cut-off. A data-driven weighting recovers the recall but gives a user less to read, and a deployment can choose between them knowing the price.

Explanations differ by component. The post and conversation detectors are neural models, so token-level attributions are suitable: LIME and leave-one-out attribution for the post detector, and SHAP and LIME for the conversation detector. The employer verifier is a fusion of named evidence sources, so the honest explanation is the evidence itself and its per-layer share of the score, which can be checked by the user on the official registers. The paper does not claim that one form is better, only that each fits its component.

The employer study shows that the position of the risk thresholds decides which error occurs. The deployed bands

keep the Low band clean, and the validation-selected bands catch more fraud. Which is right depends on whether a missed scam or a wrongly flagged genuine employer is worse for users, and on a platform's tolerance of complaints from employers.

Conversation text and screenshots are personal data, so a deployment should process them only on request, keep them no longer than needed, and show users the evidence and not only a verdict, since a false accusation harms a genuine employer.

The validity of these findings is limited in three ways. The labels of the post detector and the conversation samples were not produced by independent annotators, so the measured accuracy may differ from agreement with human judgement. The employer set was assembled for this study and its fraud class comes from reported schemes, so unreported or newly created scams may behave differently. The Sri Lankan setting also limits how far the numbers carry to other countries, whose registers, languages and scam styles differ.

## VIII. Limitations and Conclusion

The components were evaluated on separate datasets, so the reported metrics describe each component and not the joint accuracy of the whole pipeline. The labels of the post detector were assigned by an automated labeller, and the conversation data include only 322 real collected messages, so both results describe controlled offline performance and not live traffic. Some employer labels come from unverified social-media reports, and some feature values were randomised at fixed rates to create overlap between the classes. The employer test split has only 300 employers, so its intervals are wide, and the deployed scorer uses capped points with graded registry points, so it is not identical to Eq. (1). EWM weights depend on the candidate set, and the weights used in the TOPSIS step are not tuned. Component-level results of the post and conversation detectors also appear in a manuscript under review [26]; the system-level integration, the employer study and the ranking method are new here.

In conclusion, this paper presented an architecture in which post, employer and conversation detectors share a risk-and-explanation contract, and a ranking method that uses EWM and TOPSIS to rank jobs by relevance and safety, which lowers the average risk of the Top-10 list from 0.101 to 0.090 without losing relevance. The post and conversation detectors reach 88.5% and 85.3% accuracy, and the employer study shows that explainable weights keep fraud out of the Low band while catching 78.1% of fraudulent employers, 15 points below a data-driven weighting.

Future work has three steps. First, a quantitative end-to-end test of the whole pipeline on one common set of labelled jobs, so that the accuracy of the combined system can be reported. Second, a larger and independently labelled employer test set, human-annotated labels for the post detector, and more real conversations. Third, wider language coverage, starting with Tamil in addition to Sinhala and English, together with regular retraining as scam wording changes.

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